

Project Proposal

Student Debt vs. Wealth Milestones

Group 1: Andrew Grant, Anna Janvrin, Vittoria Quaranta, Dimitri Dorsey, Roberto Rendon

March 11, 2026

1 Introduction

1.1 Project goal

Our goal is to find out how student debt affects future financial milestones. These milestones are: buying a house, investing in a Roth IRA, age at retirement, and total investments.

1.2 Motivation

Many Students are going to college and accepting the financial burden that comes with it. We aim to see whether college is worth it by analyzing future financial results. Several studies have been conducted on whether college is worth it. This project aims to see whether student debt is worth it.

1.3 Background and related work

General Context

In recent years, student debt has become an increasing burden for many students. In fact, the US News and World Report estimates that in 2024, 56% of graduates had taken out student loans. It is likely that such significant debt could correlate to delayed adulthood financial milestones. So, while a degree may increase earnings later on, are the increased earnings worth the financial burden the debt carries? What could someone have done without having to worry about that debt? Could they retire earlier? Buy a house earlier? This project dives into the long term impact of the initial student debt levels on the timing of key financial milestones. We are looking for when an individual's worth increases because they can achieve milestones such as investments and home ownership.

Relevant Questions:

- For every additional \$10,000 in student debt, how much longer does it take until one purchases a home of value (INSERT RANGE)?
- Is there a statistically significant correlation between high undergraduate debt levels and the retirement age?
- How does the amount of student debt affect when a person funds a ROTH IRA account?
- Does student debt lead to a delay in investing at least \$30,000?

Summary of existing approaches:

There has been a few relevant studies that conducted research similar to our question. Fidelity did a study on student loans and future financial achievements. Fidelity is a retirement plan provider so their data comprises data from millions of 401k and 403b accounts. Fidelity also has a student debt tool where a person's debt data is transferred. For their other measures, they used their annual surveys. They then used models that utilized cross-sectional surveys. Another study was done by the Federal Reserve that analyzed how student debt affects consumer spending. They compared consumer spending data across many ZIP codes with varying levels of exposure to student loan debt. They analyzed credit and debit transaction data. They applied time trends and then calculated that every 10k in debt led to a significant reduction in household consumption. Another study we found was by the TIAA Institute. They did a financial wellness survey of 2001 student loan borrowers in 2024. They explored how borrowers make decisions and how their financial behaviors were affected by using a Personal Finance Index, which was a survey of 28 questions about financial concepts. This led to findings about how specific gaps in financial understanding cause financial delays from student debt.

2 Intellectual merits and broader impacts

2.1 Central question

The problem is that students are going to college and acquiring debt. This debt could prevent the students from reaching their future goals financially. There is a heavy debate on whether students should go to college and incur that debt in the first place. So we are asking, how does student debt affect future financial milestones?

2.2 Specific aims and assumptions

Our objectives:

- Through analyzing long-term household data, mortgage loans, and student debt-income, our goal is to discover how an additional \$10,000 in debt affects how long it takes for that person to purchase a home.
- By using PCA and statistical analysis, our goal is to see if there is a statistically significant correlation between high undergraduate debt levels and retirement age.
- Through using the financial history of individuals, our goal is to see if there is a connection between the amount of student debt one requires and when that person opens a Roth IRA account.
- Our last goal is to identify if student debt leads to a delay in investing 30,000 through looking at long-term financial data.

Our assumptions:

- time period: 2004-2024 -For the retirement age, we would calculate what amount of saving is needed to retire using the $4\text{-total needed} = \text{desired annual income}/4$ - most people need 70-80
- 10,000 in debt, and 30,000 in investments is adjusted for inflation
- For the Roth IRA and Investment, we will assume that the market returns are constant at a standardized return of 7

- Assuming that the criteria for homeownership remain relatively stable over our data period
- Assume that the delay in milestones is from student debt and not other factors like lifestyle choices
- other types of debt are negligible in the delays for milestones

2.3 Planned deliverables

What will success look like?

- Once our data is analyzed, and we detect any possible patterns, we can judge our success by seeing if our model can quantify how much each person who has debt is set back from our defined financial milestones, depending on the amount of debt they take on, through the use of regression.

3 Data and analysis

3.1 Datasets and processing

Describe potential datasets or repositories, data formats, and any anticipated cleaning, merging, or assumptions.

- The University of Michigan's Panel Study of Income Dynamics is the world's longest running longitudinal household survey. It has household data on a variety of economic and financial factors, including amount in retirement, mortgage debt, consumer debt, etc. While the dataset requires significant merging across waves, unique household IDs are provided to facilitate linking records over time.
- The 2022 Survey of Consumer Finances (SCF) is a Federal Reserve Dataset providing a comprehensive snapshot of U.S. household assets and debts. We access it via UC Berkeley's SDA tool, which allows us to instantly filter and analyze complex variables through a web-based interface.
- The College Scorecard raw data is public data acquired from the US Department of Education. This includes information on cumulative debt at graduation and earnings one year after graduation. It allows us to compare debt and early earnings. There are data files from 1996-97 through 2022-23 containing aggregate data for each institution. It includes information on institutional characteristics, enrollment, student aid, costs, and student outcomes. These can help us analyze the financial factors of going to college.

3.2 Modeling and analysis

Describe the analysis direction you plan to pursue and the program(s) you expect to use. Briefly explain why this direction fits your question and data.

- We will need to perform PCA through Colab to establish which factors(interest rates, debt size, income) explain the most variance in financial delay. Student debt can have a lot of related pressures, like interest rates, income, monthly payments, total balance, etc. PCA can allow us to combine variables into a single factor to analyze.

- We can then use coding on Colab to run a regression on the variables to find if there exists a correlation. Furthermore, we can create graphs to show the relationship between debt and financial milestones. Graphs could be created through RStudio or Collab. When we are able to see a correlation between variables, we can attempt to identify the relationship between them. If we can find the relationship, we can answer our question about how student debt affects future financial aspects.

4 Work plan and feasibility

Our group has three main questions: how debt affects investment, retirement, and buying a house. We have decided to split our three questions among the people in our group. Andrew and Antonio will research how student debt impacts buying a house. Anna and Dimitri will analyze how student debt affects retirement age. Vittoria will see how student debt affects having \$30,000 in investments. Roberto will research how student debt impacts when someone adds funds to a Roth IRA. First, we will need to clean the data to break down which variables and factors we need to perform our analysis. This starts with investigating the database and determining which variables will be useful. We will then need to clean the code and unify it by family across years. Then, we import our datasets into Colab, where we can collaboratively code the data to perform analysis that will show relationships between variables. We will then need to create visual aids that can assist in the findings of our data. Once we finalize our visuals and numerical findings, we will compile them into a PowerPoint. We believe this work is feasible within 5-6 weeks because the first two weeks, we can finetune our data that can be imported to Colab. The two weeks after that we can run the data analysis through the division of work we have outlined. The final weeks, we will finalize our findings into a formal presentation.

List of everything that will need to get done:

- data-cleaning: significantly clean the data into a CSV or file that can be used to run PCA on and other data analysis tools; make sure the data can be analyzed
- import data into Google Colab to begin coding
- Run PCA, Linear Regressions, or other data analysis tools to answer our question about the relationship between student debt and financial milestones
- create graphs to show our data findings and descriptions to go with the graphs
- condense our findings into a PowerPoint presentation that can be 15 minutes long

Weeks 1–2: Focus exclusively on cleaning and merging the PSID and College Scorecard data. This is the most labor-intensive phase.

Weeks 3–4: Import cleaned data into Google Colab. Run PCA to identify latent debt factors and perform regressions against our four milestones (Home, Retirement, Roth IRA, \$30k Investment).

Weeks 5–6: Finalize visualizations, interpret p-values, and condense the findings into the 15-minute presentation